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Editorial Board

Quantum – the open journal for quantum science

Dear Editors,

We are pleased to submit our manuscript entitled “**A Reproducible Precision-Aware Workflow for Unanchored Approximate Mutually Unbiased Bases in Dimension Six**” for consideration for publication in *Quantum*.

Our work addresses a classical, longstanding challenge in quantum information theory and finite-dimensional Hilbert space geometry: the existence of a complete set of seven Mutually Unbiased Bases (MUBs) in dimension six ($d = 6$). We approach this from a mathematical software perspective by introducing an unanchored optimization workflow that utilizes automatic differentiation and Lie-algebra exponentiation to search for approximate configurations.

We believe our manuscript is an excellent fit for *Quantum*, as it provides rigorous numerical evidence for a problem central to quantum information theory, while introducing precision-aware computational tools for high-dimensional quantum combinatorial design searches:

- **Unanchored AMUB Formulation:** We study a fully unanchored optimization problem, exposing structural patterns and showing numerical evidence of a transition to fully defective configurations beginning at five bases in dimension six.
- **Hardware-Accelerated Exponentiation Layer:** To support GPU acceleration natively, we developed a highly portable complex matrix Taylor-series exponentiation layer, bypassing backend limitations while maintaining strict truncation error bounds.
- **Precision-Aware Landscape Analysis:** We conduct a thorough analysis contrasting double-precision (`complex128`) reference runs with single-precision (`complex64`) implementations, demonstrating how precision impacts near-exact pair classifications and optimization basin selection.
- **Reproducible Software Workflow:** Our methodology is packaged as a modern, reproducible workflow leveraging Python, PyTorch, and interactive notebooks. All optimization seeds, overlap tensors, diagnostics, and resulting basis matrices are fully archived and publicly exportable, allowing independent verification by the quantum computing community.

This manuscript has not been published elsewhere and is not under consideration by any other journal. All authors have read and approved the final manuscript for submission.

Thank you very much for your time and for considering our work. We look forward to receiving

the reviews.

Sincerely yours,

Abdul Fatah

On behalf of the authors:

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